

**National University of Computer and Emerging Sciences**



## **Assignment No. 5**

**Generative AI**

AI4009

Spring 2026

## Submission Instructions:

- Submit the complete .ipynb file, which should be named AI\_ASS01\_XXF\_YYYY, where XX represents the batch and YYYY represents the roll number.
- Submit a Word file containing the GitHub repository link, as well as the Medium and LinkedIn post links
- This is a group assignment (max 2 members allowed).
- Any submission not following the submission instructions will not be evaluated.
- Explore the Datasets completely before starting to code.
- You can explore more datasets if needed and must explain your reasoning of using them but there is a constraint attached datasets must need to be used.
- **Submit your own work. Accept your current limitations. Grow from your mistakes.**
- **Plagiarism is strictly prohibited and lead to Straight Zero (^\_^).**

## Question: Vision Language Models (VLM) using QLoRA Fine-Tuning for Document-to-Markdown Generation

**Objective:** The objective of this assignment is to fine-tune a Vision Language Model (VLM) using QLoRA for the task of converting document images into structured Markdown documentation.

You will:

- Understand multimodal learning (image + text)
- Perform parameter-efficient fine-tuning
- Adapt a pretrained VLM to document understanding
- Generate Markdown outputs from unseen document images

### 1. Environment Setup

- **Platform:** <https://www.kaggle.com/>
- **Accelerator:** GPU T4 x2 (Dual GPU)
- **Framework:** PyTorch + Hugging Face Transformers + PEFT
- **Dataset:**
  - **Nougat Training Dataset Example:** <https://www.kaggle.com/datasets/zphilip/nougat-training-dataset-example>

### 2. Model Architecture

#### ➤ Vision Language Model

Use the pretrained **Qwen2-VL-2B-Instruct**.

**Model page:** <https://huggingface.co/Qwen/Qwen2-VL-2B-Instruct>

### 3. Fine-Tuning Technique

#### ➤ QLoRA (Quantized Low-Rank Adaptation)

Instead of full fine tuning:

- load the base model in 4-bit quantized form
- Freeze original pretrained weights
- Train only lightweight LoRA adapters

### 4. Implementation Tasks

#### Part 1: Dataset Exploration

1. Load the Nougat dataset
2. Inspect image – markdown pairs
3. Understand visual layout and document structure
4. Select a subset of samples or full dataset for training

#### Part 2: Data Preparation

Convert the dataset into ChatML format. Each sample should contain:

1. Image
2. Instruction prompt
3. Markdown target

#### Part 3: Dataset Subsetting

Recommended split:

- 80% training
- 20% validation

Or use can use your specified settings

#### Part 4: QLoRA Fine-Tuning

Fine-tune the model using:

- 4-bit quantization
- LoRA adapters
- 2–5 epochs

#### Required Training Techniques (to fit Kaggle T4×2)

- batch size: 1–2
- gradient accumulation: 4–8
- learning rate: 1e-4 to 2e-4
- LoRA rank: 8 or 16
- image resolution: 512–768 px
- or your recommended settings

#### Part 5: Markdown Generation (CORE TASK)

For each validation image:

1. Run the fine-tuned model

2. Generate Markdown output
3. Compare with ground-truth

## Part 6: Testing Task

Generate Markdown Outputs for:

- 3 training images
- 3 unseen document images

## 5. Visualization Module

Display:

- Input image
- Ground truth markdown
- Generated markdown

## Deliverables

### 1. Jupyter Notebook:

- Complete PyTorch implementation.
- Clean, well-structured code

### 2. Fine Tuning Results

- Training loss
- Validation loss
- Generated markdown outputs
- Comparison (Actual vs Predicted)

### 3. Screenshots

- Output predictions
- Visualization results

### 4. App Deployment:

- Build a Gradio or Streamlit App that should:
  - Upload image
  - Generate Markdown

## Bonus Tasks (Optional)

- Compare 2 epochs vs 5 epochs
- Compare different prompt styles
- Compare zero-shot vs fine-tuned outputs